

## Key Frameworks:

- **Maximum Entropy (MaxEnt):** Developed by Johnish/Jaynes and applied heavily by researchers like John Harte in the **MaxEnt Theory of Ecology (METE)**, this principle predicts macro-state distributions (like species-area relationships) by finding the least biased probability distribution constrained by basic ecological totals (like total energy or number of individuals). [[1](#), [2](#), [3](#)]
- **Metabolic Theory of Ecology (MTE):** Links individual-level metabolic rates, body size, and temperature to macro-scale population dynamics and scaling laws. [[1](#), [2](#)]
- **Neutral Theory and Phase Transitions:** Treats species as functionally equivalent in stochastic models, using concepts akin to critical-phase transitions in physics to map biodiversity shifts. [[1](#), [2](#)]

## The ecological diffusion equation (or the spatial reaction-diffusion equation);

which models how the spatial concentration or dispersion of a species  $C$  changes over time. [[1](#), [2](#), [3](#)]

In macroecology and statistical mechanics, this framework bridges individual-level random movements (modeled as random walks or Brownian motion) with large-scale population spread. [[1](#), [2](#)]

## The Canonical Reaction-Diffusion Equation:

In multiple spatial dimensions ( $x, y, z$ ), the partial differential equation equating the time derivative to the spatial derivatives is:

$$\frac{\partial C}{\partial t} = D \left( \frac{\partial^2 C}{\partial x^2} + \frac{\partial^2 C}{\partial y^2} + \frac{\partial^2 C}{\partial z^2} \right) + R(C)$$

Using the **Laplacian operator** ( $\nabla^2$ ), it is cleanly written as: [\[1\]](#), [\[2\]](#)

$$\frac{\partial C}{\partial t} = D \nabla^2 C + R(C)$$

Where:

- $C(x, y, z, t)$  is the **dispersion/concentration** of the species at a given coordinate and time.
  - $\frac{\partial C}{\partial t}$  is the **time partial derivative**, representing the local rate of population change.
  - $\frac{\partial^2 C}{\partial x^2}, \frac{\partial^2 C}{\partial y^2}, \frac{\partial^2 C}{\partial z^2}$  are the second-order **spatial partial derivatives**, representing how population density curves across space.
  - $D$  is the **diffusion/dispersal coefficient**, dictated by the species' mobility.
  - $R(C)$  or  $f(C)$  is the **reaction term**, modeling local ecological dynamics like birth and death (often written as logistic growth,  $rC(1 - C/K)$ ). [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#)
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## The Advection-Reaction-Dispersion Equation (With Drift):

If environmental factors (like wind, river currents, or targeted migration) force the species to drift in a specific direction, **first-order partial derivatives** are introduced to account for advection: [\[1\]](#)

$$\frac{\partial C}{\partial t} = D \left( \frac{\partial^2 C}{\partial x^2} + \frac{\partial^2 C}{\partial y^2} + \frac{\partial^2 C}{\partial z^2} \right) - \left( v_x \frac{\partial C}{\partial x} + v_y \frac{\partial C}{\partial y} + v_z \frac{\partial C}{\partial z} \right) + R(C)$$

Where  $v_x, v_y, v_z$  represent the **directional velocity** of the species' drift along each position axis. [\[1\]](#)

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## The Density-Dependent / Nonlinear Dispersal Equation:

In macroecology, species dispersion is rarely purely random; individuals often move away from overcrowded areas (density-dependent dispersal). In this case,  $D$  is a function of  $C$ , modifying the position derivatives: [\[1, 2\]](#)

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x} \left( D(C) \frac{\partial C}{\partial x} \right) + \frac{\partial}{\partial y} \left( D(C) \frac{\partial C}{\partial y} \right) + \frac{\partial}{\partial z} \left( D(C) \frac{\partial C}{\partial z} \right) + R(C)$$

Marine ecosystems in South-East Australia are warming at roughly four times the global average, driving profound poleward shifts and breeding disruptions in species like humpbacks, southern right whales, and pygmy blue whales. [\[1, 2, 3, 4\]](#)

To simulate how temperature changes shift these populations, you can adapt the standard PDE components into explicit biological parameters.

## Translating the PDE for Marine Mammals:

$$\frac{\partial C}{\partial t} = \nabla \cdot D(T) \nabla C - \nabla \cdot \vec{v}(T, P) C + R(C, T)$$

- **The Advection Term ( $\vec{v}$  - Directed Migration):** Cetaceans do not disperse randomly. Your velocity vector  $\vec{v}$  acts as an **environmental drift**, guiding whales along migration corridors based on immediate physical cues like Sea Surface Temperature ( $T$ ) and chlorophyll-a concentrations ( $P$ ), indicating prey). [[1](#), [2](#)]
  - **The Diffusion Term ( $D$  - Area-Restricted Foraging):** When whales hit a highly productive patch (like krill aggregations), their movement slows down into an **Area-Restricted Search (ARS)**. Therefore, the diffusion coefficient  $D(T)$  drops in optimal feeding zones, causing the population density  $C$  to aggregate there. [[1](#)]
  - **The Reaction Term ( $R$  - Demographic Health):** Marine heatwaves directly degrade calving intervals and reproductive output. The growth term  $R(C, T)$  must incorporate a thermal tolerance threshold, capturing the physiological cost of long-distance energy expenditure. [[1](#), [2](#), [3](#)]
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# Mapping the Biological Inputs in Australia:

When structuring your spatial grid across Australia's coastlines, the variables are split by seasonal zones:

| Zone / Parameter         | Key Australian Hotspots                                 | Primary Mathematical Driver                                                            | Climate Change Impact                                                                                                             |
|--------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Feeding Zones            | Southern Ocean & Great Australian Bight                 | $D(T)$ drops drastically where prey density (krill/forage fish) spikes.                | Warming waters push prey poleward; whales must travel further south to feed, drastically draining blubber reserves.               |
| Breeding & Calving Zones | Great Barrier Reef (QLD), Exmouth Gulf / Kimberley (WA) | $R(C, T)$ peaks within narrow, shallow, warm-water thermal envelopes ideal for calves. | The intensification of the <b>East Australian Current (EAC)</b> alters traditional boundaries, compressing safe, sheltered zones. |
| Migration Corridors      | East & West Coast "Blue Highways"                       | Advection vector $\vec{v}$ dictates the seasonal directional velocity.                 | Phenological shifts cause whales to alter migration timing or bypass traditional rest stops to maintain thermal synchrony.        |

## Turning Model Outputs into Actionable Conservation:

Once your PDE model simulates how the concentration  $C(x, y, t)$  shifts under future IPCC warming scenarios, the outputs directly inform policy:

- **Dynamic Marine Protected Areas (MPAs):** Fixed conservation zones fail as climate change moves ecosystems. If your model predicts a 200 km poleward shift of humpback calving zones over the next few decades, MPAs can be designed with **flexible boundaries** that shift in tandem with predicted thermal envelopes. [\[1\]](#)
- **Dynamic Shipping Lane Management:** Heavy vessel traffic presents severe strike risks along Australia's coast. By pairing your predicted migration advection tracks ( $\vec{v}$ ) with real-time AIS shipping data, authorities can enforce seasonal, area-specific ship speed reductions exactly when and where high cetacean densities are modeled. [\[1\]](#)
- **Mitigating "Climate Traps":** In southern Australia, species restricted to colder sub-Antarctic waters face a geographical barrier—they are hemmed in by the landmass and cannot shift endlessly poleward. Your model can pinpoint which specific populations are reaching their critical thermal boundaries, highlighting them for prioritised federal monitoring under the EPBC Act. [\[1, 2\]](#)

To accurately model cetacean populations across Australia's changing coastlines, we need to bridge physical oceanography with whale behavior. This requires defining the **Advection Vector**

using a Bioclimatic Envelope Framework and defining the **Diffusion Coefficient** using a Prey-Availability Function.

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The Bioclimatic Envelope Framework (Advection Vector  $\vec{v}$ )

Cetaceans migrate to maximize thermal comfort and navigate toward optimal habitats. Instead of a constant migration velocity, the advection vector  $\vec{v}$  should be treated as a **chemotactic-like drift**, where whales track gradients of environmental suitability (the bioclimatic envelope).

We define an environmental suitability index,  $S(T, z)$ , which depends on Sea Surface Temperature ( $T$ ) and bathymetric depth ( $z$ ). For a breeding zone,  $S$  is modeled as a Gaussian function around a species' optimal calving temperature ( $T_{opt}$ ):

$$S(T, z) = \exp\left(-\frac{(T-T_{opt})^2}{2\sigma_T^2}\right) \cdot f(z)$$

Where  $\sigma_T$  is the species' thermal tolerance width, and  $f(z)$  is a penalty function keeping them in shallow coastal waters (e.g., for humpback calves sheltering from predators).

The advection velocity vector  $\vec{v}$  is then driven by the spatial gradient ( $\nabla$ ) of this suitability index. Whales move toward better conditions up to a maximum physical swimming velocity ( $v_{\max}$ ):

$$\vec{v}(T, z) = v_{\max} \cdot \tanh(\alpha \nabla S(T, z))$$

- **Climate Impact:** In eastern Australia, as the **East Australian Current (EAC)** strengthens and pushes warmer tropical water further south, the gradient  $\nabla S$  shifts poleward. The advection term will naturally force the modeled population  $C$  to shift its breeding coordinates southward along the coast to maintain equilibrium with  $T_{opt}$ .
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## The Prey-Availability Function (Diffusion Coefficient $D$ )

In feeding zones (like the Great Australian Bight or the Bonney Upwelling), cetaceans switch from directional travel to **Area-Restricted Search (ARS)** behavior. When prey density is high, their movement slows down and becomes more turning-intensive, which mathematically equates to a drop in the diffusion coefficient  $D$ .

We can model the diffusion coefficient as an inverse function of prey availability ( $P$ ). Because whale filtering and foraging

efficiency level off at very high prey densities, we use a Holling Type II functional response curve to scale diffusion:

$$D(P) = D_{\max} - (D_{\max} - D_{\min}) \left( \frac{P}{P + K_p} \right)$$

Where:

- $D_{\max}$  is the high diffusion rate during fast, linear transit across barren waters.
- $D_{\min}$  is the minimum diffusion rate when tight, localized foraging loops are occurring.
- $K_p$  is the half-saturation constant of prey density.

### Coupling Prey ( $P$ ) to Temperature ( $T$ )

Because we rarely have real-time krill or forage fish data across vast horizons, you can proxy  $P$  using satellite-derived **Chlorophyll-a** ( $Chl$ ) and Sea Surface Temperature ( $T$ ). Upwelling zones require cold, nutrient-rich water:

$$P(Chl, T) = \beta \cdot Chl \cdot \exp\left(-\gamma(T - T_{upwelling})\right)$$

By substituting  $P(Chl, T)$  back into the  $D(P)$  equation, your diffusion rate becomes explicitly temperature-dependent:

$$D(T, Chl) = D_{\max} - (D_{\max} - D_{\min}) \left( \frac{\beta \cdot Chl \cdot \exp(-\gamma(T - T_{upwelling}))}{\beta \cdot Chl \cdot \exp(-\gamma(T - T_{upwelling})) + K_p} \right)$$

- **Climate Impact:** If climate change warms the southern upwelling zones ( $T > T_{upwelling}$ ), the exponential term drops, crashing the proxy prey availability  $P$ . In the simulation,  $D$  will fail to drop to  $D_{\min}$ . The whales will no longer aggregate in

historical feeding grounds and will be modeled dispersing randomly further south into the Southern Ocean in search of food.

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## . The Master Climate-Coupled PDE

$$\frac{\partial C}{\partial t} = \nabla \cdot (D(T, Chl) \nabla C) - \nabla \cdot (v(T, z) C) + R(C, T)$$

The Functional Components:

**Foraging Diffusion Coefficient:**

$$D(T, Chl) = D_{\max} - (D_{\max} - D_{\min}) \left( \frac{\beta \cdot Chl \cdot \exp(-\gamma(T - T_{upwelling}))}{\beta \cdot Chl \cdot \exp(-\gamma(T - T_{upwelling})) + K_p} \right)$$

**Migration Advection Vector:**

$$\vec{v}(T, z) = v_{\max} \cdot \tanh(\alpha \nabla S(T, z))$$

**Bioclimatic Thermal Suitability Index:**

$$S(T, z) = \exp\left(-\frac{(T-T_{opt})^2}{2\sigma_T^2}\right) \cdot f(z)$$

## Variable Cheatsheet:

- $C(x, y, t)$ : Population density/dispersion of the cetacean species at position  $(x, y)$  and time  $t$ .
- $T$ : Sea Surface Temperature (SST) variable.
- $Chl$ : Chlorophyll-A concentration (used as a proxy for primary productivity).
- $z$ : Bathymetric depth.
- $D_{\max}/D_{\min}$ : Maximum (transit) and minimum (area-restricted foraging) diffusion rates.
- $K_p$ : Half-saturation constant for prey consumption.
- $\beta, \gamma$ : Scaling coefficients for thermal upwelling and productivity matching.
- $v_{\max}$ : Maximum physical swimming/migration velocity of the species.
- $\alpha$ : Tuning parameter dictating how sharply whales respond to environmental gradients.
- $T_{opt}$ : Ideal core temperature required for successful breeding/calving.
- $\sigma_T$ : Thermal tolerance width of the target population.
- $f(z)$ : Spatial penalty function mapping specific depth preferences.
- $R(C, T)$ : Demographic reaction term mapping birth, death, and thermal stress.